



RECREO HUB

Computer Programming Lab Project Report

Bahria University Karachi Campus

COMPUTER PROGRAMMING -LAB PROJECT REPORT

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Semester: First

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Report on "Recreohub" Project

Overview

The "Recreohub" project implements an interactive system that integrates multiple facilities such as a Cinema House, a Coffee Shop, Indoor Games for members, and Arcade Games. The system utilizes a structured menu-driven interface to manage different operations like bookings, orders, and game sessions. Below is a detailed analysis of its components:

Main Functionalities

1. **Cinema House:**
 - Allows users to select movie genres (Action, Horror, Drama) and specific movies within each genre.
 - Enables seat booking with validation for availability.
 - Ticket price recording is implemented for revenue tracking.
 - Movies are organized using a structured Movie array.
 2. **Coffee Shop:**
 - Offers Iced Coffees, Starters, and Desserts.
 - Customization options like blends, extra sauces, and cheese are available.
 - Orders are recorded and displayed for confirmation.
 - Revenue from coffee shop sales is logged for earnings computation.
 3. **Indoor Games (Members Only):**
 - Games include Snooker, Table Tennis, and Squash.
 - Access is restricted to validated members through a login system.
 - Gameplay involves score tracking with randomized results.
 4. **Arcade Games:**
 - Includes Hangman, Guess the Number, and Tic Tac Toe.
 - Offers simple interactive gameplay with random or logical outcomes.
 5. **Purchases and Earnings:**
 - Operators can log in to view total earnings from Cinema House and Coffee Shop.
 - Individual and combined earnings are calculated using file-based storage.
-

Code Highlights

1. **Structured Programming:**
 - Functions are modular, focusing on individual tasks like order handling, game execution, and earnings computation.
 - Structs (Coffee, Starter, Dessert, Movie) ensure organized data representation.
 2. **File Management:**
 - Earnings from the Cinema House and Coffee Shop are stored in respective text files for persistent tracking.
 - Uses ofstream and ifstream for appending and reading data.
 3. **Interactive Console Output:**
 - Uses ANSI escape codes for colored text output to enhance user experience.
 - Menus and outputs are formatted neatly with separators and alignments.
 4. **User Authentication:**
 - Two levels of authentication:
 - Operators for viewing earnings.
 - Members for accessing Indoor Games.
 5. **Randomized Gameplay:**
 - Games like Snooker, Table Tennis, and Squash use random numbers for scoring.
 - Hangman employs random word selection from a predefined array.
-

Strengths

1. **Rich User Experience:**
 - Colorful and formatted menus.
 - Diverse options catering to entertainment and food preferences.
 2. **Revenue Tracking:**
 - Systematic logging and viewing of earnings.
 3. **Engaging Gameplay:**
 - Variety of games ensures user engagement.
 4. **Modularity:**
 - Clear segregation of functionality.
-

Potential Extensions

1. **Online Integration:**
 - Connect the system to an online database for centralized data management.
 2. **User Profiles:**
 - Allow members to have personalized accounts and saved preferences.
 3. **Enhanced Gameplay:**
 - Introduce more games with advanced logic or multiplayer options.
 4. **Dynamic Pricing:**
 - Add a feature to manage and adjust item prices dynamically.
-

This project is an excellent demonstration of combining functional programming with user interaction. Implementing the suggested improvements could make it more robust and scalable.

Screen Shots (SS) of Code:

Variable, Pointers & Arrays & 2D Arrays:

```
Movie* moviepointer = movies;
const int rows = 5;
const int seatsperrow = 6;
char seats[rows][seatsperrow];
char (*seatspntr)[seatsperrow] = seats;
```

```
string order[7][2];
char blendchoice, extrasauce, extracheese, orderc;
int option, exit, billamount = 0;
string order1;
Coffee coffees[6];
Starter starters[6];
Dessert desserts[4];
```

Structure:

```
struct Coffee
{
    string name;
    int price = 0;
};

struct Starter
{
    string name;
    int price = 0;
};

struct Dessert
{
    string name;
    int price = 0;
};

struct Movie
{
    string name;
    string duration;
    int reldate = 0;
    string date;
};
```

Functions(Recurssion & Overloading Included):

```
int Module(char cofchoice, int blend, char extrashot, char sofchoice, char dofchoice); // coffeeshop
int Module(int exit, int choice); // cinemahouse
void DisplayTotalEarnings();
void RecordCinemaEarnings(double TicketPrice);
void RecordCoffeeEarnings(double BillAmount);
bool ValidateMembership();
void PlaySnooker();
void PlayTableTennis();
void PlaySquash();
int GetRandomNumber(int min, int max);
void DisplayWord(const string& word, const bool guessed[]);
void PlayHangman();
void PlayNumberGuessingGame();
void DisplayBoard(const char board[9]);
char CheckWinner(const char board[9]);
bool IsBoardFull(const char board[9]);
void PlayTicTacToe();
void RecreoHubMenu();
```

File handling

```
//Cinema house earning fn
void RecordCinemaEarnings(double ticketPrice)
{
    ofstream outFile("CinemaHouseEarnings.txt", ios::app);
    if (outFile.is_open())
    {
        outFile << ticketPrice << endl;
        outFile.close();
    }
    else
    {
        cerr << " Error: Unable to open CinemaHouseEarnings.txt " << endl << "\n";
    }
}

void RecordCoffeeEarnings(double billamount) {
    ofstream outFile("CoffeeShopEarnings.txt", ios::app);
    if (outFile.is_open())
    {
        outFile << billamount << endl;
        outFile.close();
    }
    else
    {
        cerr << " Error: Unable to open CoffeeShopEarnings.txt " << endl << "\n";
    }
}
```


CODE:

```
#include <iostream>

#include <ctime>

#include <iomanip>

#include <fstream>

#include <string>

using namespace std;


int Module(char cofchoice, int blend, char extrashot, char sofchoice, char dofchoice); //
coffeeshop

int Module(int exit, int choice); // cinemahouse

void DisplayTotalEarnings();

void RecordCinemaEarnings(double TicketPrice);

void RecordCoffeeEarnings(double BillAmount);

bool ValidateMembership();

void PlaySnooker();

void PlayTableTennis();

void PlaySquash();

int GetRandomNumber(int min, int max);

void DisplayWord(const string& word, const bool guessed[]);

void PlayHangman();

void PlayNumberGuessingGame();

void DisplayBoard(const char board[9]);

char CheckWinner(const char board[9]);
```

```
bool IsBoardFull(const char board[9]);  
  
void PlayTicTacToe();  
  
void RecreoHubMenu();  
  
void Exit_Menu();  
  
double cinemaEarnings, coffeeEarnings;
```

```
void SetColor(int textColor)  
{  
    cout << "\033[" << textColor << "m";  
}  
  
void ResetColor() { cout << "\033[0m"; }
```

```
struct Coffee  
{  
    string name;  
    int price = 0;  
};
```

```
struct Starter  
{  
    string name;  
    int price = 0;  
};
```

```
struct Dessert
```

```
{  
  
    string name;  
  
    int price = 0;  
  
};
```

```
struct Movie
```

```
{  
  
    string name;  
  
    string duration;  
  
    int reldate = 0;  
  
    string date;  
  
};
```

```
int main()
```

```
{  
  
    int choice, exit = 0, blend = 0, IndoorGameChoice = 0, ArcadeGameChoice = 0;  
  
    char cofchoice = 0, extrashot = 0, sofchoice = 0, dofchoice = 0;  
  
    SetColor(90);  
  
    cout << setfill('~') << setw(50) << "~" << endl;  
  
    cout << " " << char(4) << " " << char(4) << " " << char(4) << " Welcome to  
Recreohub " << char(4) << " " << char(4) << " " << char(4) << endl;  
  
    cout << setfill('~') << setw(50) << "~" << endl << "\n";
```

```

do
{
    SetColor(95);

    RecreoHubMenu();

    SetColor(36);

    cout << "\n" << " Kindly enter your choice: ";

    cin >> choice;

    switch (choice)
    {
    case 1:

        Module(exit, choice);

        SetColor(95);

        //RecreoHubMenu();

        SetColor(36);

        break;

    case 2:

        Module(cofchoice, blend, extrashot, sofchoice, dofchoice);

        SetColor(95);

        //RecreoHubMenu();

```

```
SetColor(36);
```

```
break;
```

```
case 3:
```

```
SetColor(91);
```

```
cout << " Indoor Games are Only for Members. " << endl << "\n";
```

```
if (!ValidateMembership())
```

```
{
```

```
    break;
```

```
}
```

```
do
```

```
{
```

```
    SetColor(32);
```

```
    cout << "\n" << setfill('*') << setw(10) << "*";
```

```
    cout << " Indoor Games. ";
```

```
    cout << setfill('*') << setw(10) << "*" << "\n";
```

```
SetColor(95);
```

```
cout << " Menu :- " << endl;
```

```
cout << " 1. Snooker. " << endl;
```

```
cout << " 2. Table Tennis. " << endl;
```

```
cout << " 3. Squash. " << endl;
```

```
cout << " 4. Exit. " << endl;
```

```

SetColor(36);

cout << "\n" << " Enter your choice. ";

cin >> IndoorGameChoice;


switch (IndoorGameChoice)
{
case 1:

    PlaySnooker();

    break;


case 2:

    PlayTableTennis();

    break;


case 3:

    PlaySquash();

    break;


case 4:

    cout << " You have chosen 4 for Exit. " << endl << "\n";

    SetColor(90);

    cout << "\n" << setfill('~') << setw(50) << "~" << endl;

    cout << "   Exiting Indoor Games...   " << endl;

```

```

        cout << setfill('~') << setw(50) << "~" << endl << "\n";

        SetColor(36);

        break;

default: cout << " Invalid Indoor Game choice has been entered. " << endl << "\n";

    }

    /*SetColor(33);

    cout << "\n" << "                To exit :- enter -1. " << endl;

    cout << "                To continue :- any number. ";

    cin >> exit;

    cout << "\n";

    */

} while (IndoorGameChoice != 4);

/*SetColor(90);

cout << "\n" << setfill('~') << setw(50) << "~" << endl;

cout << "    Exiting Indoor Games...  " << endl;

cout << setfill('~') << setw(50) << "~" << endl << "\n";

*/

break;

case 4:

    cout << " You have chosen 4 for Arcade Games. " << endl << "\n";

    while (ArcadeGameChoice != 4)

```

```

{
    SetColor(32);

    cout << setfill('*') << setw(10) << "*";

    cout << " Arcade Games. ";

    cout << setfill('*') << setw(10) << "*" << "\n";


    SetColor(95);

    cout << " Menu :- " << endl;

    cout << " 1. Hangman. " << endl;

    cout << " 2. Guess the Number. " << endl;

    cout << " 3. Tic Tac Toe. " << endl;

    cout << " 4. Exit. " << endl;


    SetColor(36);

    cout << "\n" << " Enter your choice. ";

    cin >> ArcadeGameChoice;


    switch (ArcadeGameChoice)
    {
    case 1:

        PlayHangman();

        break;

```


case 2:

```
PlayNumberGuessingGame();
```

```
break;
```

case 3:

```
PlayTicTacToe();
```

```
break;
```

case 4:

```
cout << " You have chosen 4 for Exit. " << endl << "\n";
```

```
SetColor(90);
```

```
/*cout << "\n" << setfill('~') << setw(50) << "~" << endl;
```

```
cout << "   Exiting Arcade Games...   " << endl;
```

```
cout << setfill('~') << setw(50) << "~" << endl << "\n";
```

```
exit = -1;
```

```
*/
```

```
break;
```

```
default: cout << " Invalid Arcade Game choice has been entered. " << endl <<
"\n";
```

```
SetColor(33);
```

```
cout << "\n" << "           To exit :- enter -1. " << endl;
```

```
cout << "           To continue :- any number. ";
```

```
cin >> exit;
```

```

        cout << "\n";

        /*if (exit == -1)

        {

            SetColor(90);

            cout << "\n" << setfill('~') << setw(50) << "~" << endl;

            cout << "    You have exited Arcade Games...  " << endl;

            cout << setfill('~') << setw(50) << "~" << endl << "\n";

        }

        continue;*/

    }

    /*SetColor(33);

    cout << "\n" << "                To exit :- enter -1. " << endl;

    cout << "                To continue :- any number. ";

    cin >> exit;

    cout << "\n";*/

}

SetColor(90);

cout << "\n" << setfill('~') << setw(50) << "~" << endl;

cout << "    You have exited Arcade Games...  " << endl;

cout << setfill('~') << setw(50) << "~" << endl << "\n";

break;

```

case 5:

```

        cout << " You have chosen 5 for Purchases. " << endl << "\n";

```

```

        cout << " Login to your account below. " << "\n";

        DisplayTotalEarnings();

        break;

case 6:

        cout << " You have chosen 6 for Exit. " << endl << "\n";

        Exit_Menu();

        return 0;

        break;

default: cout << " Invalid choice has been entered. " << endl << "\n";

}

/*SetColor(33);

cout << "\n" << "        To exit Recreohub :- enter -1. " << endl;

cout << "        To continue :- any number. ";

cin >> exit;

cout << "\n" << "\n";*/

} while (exit != -1);

return 0;

}

void Exit_Menu()

{

        SetColor(90);

```

```

cout << "\n" << setfill('~') << setw(50) << "~" << endl;

cout << "    Come again soon...  " << endl;

cout << setfill('~') << setw(50) << "~" << endl << "\n";

SetColor(36);

system("pause");

}

int Module(char cofchoice, int blend, char extrashot, char sofchoice, char dofchoice) //
coffeeshop

{

    string order[7][2];

    char blendchoice, extrasauce, extracheese, orderc;

    int option, exit, billamount = 0;

    string order1;

    Coffee coffees[6];

    Starter starters[6];

    Dessert desserts[4];

    coffees[0].name = " Frozen Cappucino. "; coffees[0].price = 825;

    coffees[1].name = " Iced Spanish Latte. "; coffees[1].price = 645;

    coffees[2].name = " Iced Americano. "; coffees[2].price = 495;

    coffees[3].name = " Coffee Crunch Frappe. "; coffees[3].price = 695;

    coffees[4].name = " No Blend chosen. "; coffees[4].price = 0;

    coffees[5].name = " No Extra Shot chosen. "; coffees[5].price = 0;

    starters[0].name = " Mozzarella Sticks. "; starters[0].price = 995;

```

```

starters[1].name = " Mexican Fries. "; starters[1].price = 1195;
starters[2].name = " Potato Skins. "; starters[2].price = 895;
starters[3].name = " Loaded Nachos. "; starters[3].price = 2095;
starters[4].name = " No Extra Sauce chosen. "; starters[4].price = 0;
starters[5].name = " No Extra Cheese chosen. "; starters[5].price = 0;
desserts[0].name = " Chocolate Chip Cookie "; desserts[0].price = 245;
desserts[1].name = " Waffle with Ice Cream. "; desserts[1].price = 895;
desserts[2].name = " Chocolate Heaven Cake. "; desserts[2].price = 695;
desserts[3].name = " Swiss Chocolate Brownie. "; desserts[3].price = 795;

cout << " You have chosen 2 for Coffee shop. " << endl << "\n";

do
{
    SetColor(32);

    cout << setfill('*') << setw(10) << "*";

    cout << " Coffee Shop ";

    cout << setfill('*') << setw(10) << "*" << "\n";

    SetColor(95);

    cout << " Menu :- " << endl;

    cout << " 1. Iced Coffees. " << endl;

    cout << " 2. Starters. " << endl;

    cout << " 3. Desserts. " << endl;

    cout << " 4. Exit. " << endl;

```

```

SetColor(36);

cout << "\n" << " Enter your choice. ";

cin >> option;

switch (option)
{
case 1:

    cout << " You have chosen 1 for Iced Coffees. " << endl << "\n";

    SetColor(95);

    cout << " Options :- " << endl;

    cout << " A. Frozen Cappucino. Rs 825" << endl;

    cout << " B. Iced Spanish Latte. Rs 645 " << endl;

    cout << " C. Iced Americano. Rs 495" << endl;

    cout << " D. Coffee Crunch Frappe. Rs 695 " << endl << "\n";

    SetColor(36);

    cout << " Enter your Coffee choice = ";

    cin >> cofchoice;

    cofchoice = toupper(cofchoice);

    switch (cofchoice)
    {

    case 'A':

```

```

    order[0][0] = coffees[0].name;

    order[0][1] = to_string(coffees[0].price);

    break;

case 'B':

    order[0][0] = coffees[1].name;

    order[0][1] = to_string(coffees[1].price);

    break;

case 'C':

    order[0][0] = coffees[2].name;

    order[0][1] = to_string(coffees[2].price);

    break;

case 'D':

    order[0][0] = coffees[3].name;

    order[0][1] = to_string(coffees[3].price);

    break;

default: cout << " Invalid Coffee choice has been entered. " << endl << "\n";

    SetColor(33);

    cout << "\n" << "                To exit :- enter -1. " << endl;

    cout << "                To continue :- any number. ";

    cin >> exit;

```

```

cout << "\n";

if (exit == -1)
{
    SetColor(90);

    cout << "\n" << setfill('~') << setw(50) << "~" << endl;

    cout << "    You have exited the Coffee Shop...  " << endl;

    cout << setfill('~') << setw(50) << "~" << endl << "\n";

    return 0;

}

continue;
}

```

```

SetColor(36);

cout << "\n" << " Would you like a blend? (Y/N) ";

cin >> blendchoice;

```

```

if (blendchoice == 'Y' || blendchoice == 'y')
{
    SetColor(95);

    cout << "\n" << " Pick your blend : " << endl;

    cout << " 1. Signature Blend. " << endl;

    cout << " 2. Italian Roast. " << endl;

    SetColor(36);

```



```

cout << " Choose your blend = ";

cin >> blend;


if (blend == 1)
{
    coffees[4].name = " Signature Blend. ";
    coffees[4].price = 100;
    order[1][0] = coffees[4].name;
    order[1][1] = to_string(coffees[4].price);
}

else if (blend == 2)
{
    coffees[4].name = " Italian Roast. ";
    coffees[4].price = 100;
    order[1][0] = coffees[4].name;
    order[1][1] = to_string(coffees[4].price);
}

}

else
{
    cout << " No blend has been chosen. " << endl;
    order[1][0] = coffees[4].name;
    order[1][1] = to_string(coffees[4].price);
}

```

```
cout << "\n" << " Do you want an extra shot? (Y/N) ";
```

```
cin >> extrashot;
```

```
if (extrashot == 'Y' || extrashot == 'y')
```

```
{
```

```
    coffees[5].name = " Extra Shot chosen. ";
```

```
    coffees[5].price = 175;
```

```
    order[2][0] = coffees[5].name;
```

```
    order[2][1] = to_string(coffees[5].price);
```

```
}
```

```
else
```

```
{
```

```
    cout << " No Extra Shot has been chosen. " << endl;
```

```
    order[2][0] = coffees[5].name;
```

```
    order[2][1] = to_string(coffees[5].price);
```

```
}
```

```
break;
```

```
case 2:
```

```
    cout << " You have chosen 2 for Starters. " << endl << "\n";
```

```
    SetColor(95);
```

```
    cout << " Options :- " << endl;
```

```
cout << " E. Mozzarella Sticks. Rs 995" << endl;  
cout << " F. Mexican Fries. Rs 1195 " << endl;  
cout << " G. Potato Skins. Rs 895" << endl;  
cout << " H. Loaded Nachos. Rs 2095 " << endl << "\n";
```

```
SetColor(36);  
cout << " Enter your starter choice = ";  
cin >> sofchoice;  
sofchoice = toupper(sofchoice);
```

```
switch (sofchoice)  
{  
case 'E':  
    order[3][0] = starters[0].name;  
    order[3][1] = to_string(starters[0].price);  
    break;
```

```
case 'F':  
    order[3][0] = starters[1].name;  
    order[3][1] = to_string(starters[1].price);  
    break;
```

```
case 'G':  
    order[3][0] = starters[2].name;
```

```

        order[3][1] = to_string(starters[2].price);

        break;

    case 'H':

        order[3][0] = starters[3].name;

        order[3][1] = to_string(starters[3].price);

        break;

    default: cout << " Invalid starter choice has been entered. " << endl << "\n";

        SetColor(33);

        cout << "\n" << "                To exit :- enter -1. " << endl;

        cout << "                To continue :- any number. ";

        cin >> exit;

        cout << "\n";

        if (exit == -1)

        {

            SetColor(90);

            cout << "\n" << setfill('~') << setw(50) << "~" << endl;

            cout << "    You have exited the Coffee Shop...  " << endl;

            cout << setfill('~') << setw(50) << "~" << endl << "\n";

            return 0;

        }

        continue;

    }

```

```

SetColor(36);

cout << "\n" << " Would you like Extra Sauce? (Y/N) ";

cin >> extrasauce;

if (extrasauce == 'Y' || extrasauce == 'y')
{
    starters[4].name = " Extra Sauce chosen. ";
    starters[4].price = 175;
    order[4][0] = starters[4].name;
    order[4][1] = to_string(starters[4].price);
}
else
{
    starters[4].name = " No Extra Sauce chosen. ";
    starters[4].price = 0;
    cout << " No Extra Sauce has been chosen. " << endl;
    order[4][0] = starters[4].name;
    order[4][1] = to_string(starters[4].price);
}

cout << "\n" << " Would you like Extra Cheese? (Y/N) ";

cin >> extracheese;

```

```

if (extracheese == 'Y' || extracheese == 'y')
{
    starters[5].name = " Extra Cheese chosen. ";
    starters[5].price = 225;
    order[5][0] = starters[5].name;
    order[5][1] = to_string(starters[5].price);
}
else
{
    starters[5].name = " No Extra Cheese chosen. ";
    starters[5].price = 0;
    cout << " No Extra Cheese has been chosen. " << endl;
    order[5][0] = starters[5].name;
    order[5][1] = to_string(starters[5].price);
}

break;

```

case 3:

```

cout << " You have chosen 3 for Desserts. " << endl << "\n";

SetColor(95);

cout << " Options :- " << endl;

cout << " I. Chocolate Chip Cookie. Rs 245" << endl;

cout << " J. Waffle with Ice Cream. Rs 895 " << endl;

```

```

cout << " K. Chocolate Heaven Cake. Rs 695" << endl;

cout << " L. Swiss Chocolate Brownie. Rs 795 " << endl << "\n";


SetColor(36);

cout << " Enter your dessert choice = ";

cin >> dofchoice;

dofchoice = toupper(dofchoice);


switch (dofchoice)
{
case 'I':

    order[6][0] = desserts[0].name;

    order[6][1] = to_string(desserts[0].price);

    break;


case 'J':

    order[6][0] = desserts[1].name;

    order[6][1] = to_string(desserts[1].price);

    break;


case 'K':

    order[6][0] = desserts[2].name;

    order[6][1] = to_string(desserts[2].price);

    break;

```

```

case 'L':

    order[6][0] = desserts[3].name;

    order[6][1] = to_string(desserts[3].price);

    break;

default: cout << " Invalid dessert choice has been entered. " << endl << "\n";

    SetColor(33);

    cout << "\n" << "          To exit :- enter -1. " << endl;

    cout << "          To continue :- any number. ";

    cin >> exit;

    cout << "\n";

    if (exit == -1)

    {

        SetColor(90);

        cout << "\n" << setfill('~') << setw(50) << "~" << endl;

        cout << "   You have exited the Coffee Shop...   " << endl;

        cout << setfill('~') << setw(50) << "~" << endl << "\n";

        return 0;

    }

    continue;

}

break;

```


case 4:

```
cout << " You have chosen 4 for Exit. " << endl << "\n";
```

```
SetColor(90);
```

```
cout << "\n" << setfill('~') << setw(50) << "~" << endl;
```

```
cout << "    Exiting Coffee Shop...  " << endl;
```

```
cout << setfill('~') << setw(50) << "~" << endl << "\n";
```

```
exit = -1;
```

```
continue;
```

```
break;
```

default: cout << " Invalid choice has been entered. " << endl;

```
SetColor(33);
```

```
cout << "\n" << "                To exit :- enter -1. " << endl;
```

```
cout << "                To continue :- any number. ";
```

```
cin >> exit;
```

```
cout << "\n";
```

```
if (exit == -1)
```

```
{
```

```
    SetColor(90);
```

```
    cout << "\n" << setfill('~') << setw(50) << "~" << endl;
```

```
    cout << "    You have exited the Coffee Shop...  " << endl;
```

```
    cout << setfill('~') << setw(50) << "~" << endl << "\n";
```

```
    return 0;
```

```
}
```

```

        continue;
    }

    SetColor(31);

    if (option == 1)
    {
        cout << "\n" << "\n" << " Your order :- " << endl << "\n";
        for (int i = 0; i < 3; i++)
        {
            for (int j = 0; j < 2; j++)
            {
                cout << order[i][j];

            }
            cout << endl;
        }
    }

    if (option == 2)
    {
        cout << "\n" << " Your order :- " << endl << "\n";
        for (int i = 3; i < 6; i++)

```

```

{
    for (int j = 0; j < 2; j++)
    {
        cout << order[i][j];

    }
    cout << endl;
}

}

if (option == 3)
{
    cout << "\n" << " Your order :- " << endl << "\n";
    for (int i = 6; i < 7; i++)
    {
        for (int j = 0; j < 2; j++)
        {
            cout << order[i][j];

        }
        cout << endl;
    }

}

```

```

SetColor(36);

cout << "\n" << "\n" << " Kindly confirm your order. " << endl << "\n";

cout << " Enter Y for Yes and N for No ";

cin >> orderc;

if (orderc == 'Y' || orderc == 'y')
{
    cout << " Your order is Confirmed. " << endl;

    SetColor(31);

    cout << "\n" << " Your Full Order :- " << endl;

    for (int i = 0; i < 7; i++)
    {
        for (int j = 0; j < 2; j++)
        {
            cout << order[i][j];

        }

        cout << endl;
    }

    billamount = 0;

```

```

for (int i = 0; i < 7; i++)
{
    if (!order[i][1].empty())
    {
        billamount += stoi(order[i][1]);
    }
}

RecordCoffeeEarnings(billamount);

SetColor(36);

cout << "\n" << " Total Bill is = " << billamount << endl << "\n";


SetColor(90);

cout << "\n" << setfill('~') << setw(50) << "~" << endl;

cout << "    You have exited the Coffee Shop...  " << endl;

cout << setfill('~') << setw(50) << "~" << endl << "\n";


break;

}


else if (orderc == 'N' || orderc == 'n')
{
    cout << " Your order is Not Confirmed. " << endl << "\n";
}

```

```

else
{
    cout << "\n" << " Invalid order Confirmation has been entered. " << endl << "\n";
}

billamount = 0;

for (int i = 0; i < 7; i++)
{
    if (!order[i][1].empty())
    {
        billamount += stoi(order[i][1]);
    }
}

SetColor(36);

cout << "\n" << " Total Bill is = " << billamount << endl << "\n";

SetColor(33);

cout << "\n" << "          To exit :- enter -1. " << endl;

cout << "          To continue :- any number. ";

cin >> exit;

cout << "\n";

} while (exit != -1);

return 0;

}

```

```

int Module(int exit, int choice) // cinemahouse
{
    int amchoice, hmchoice, dmchoice;

    string receipt[30];

    int row, column, tickets;

    string invalidseat = " Invalid Seat selected. "; receipt[3] = " 31st December 2024. ";

    Movie movies[12];

    movies[0].name = " Carry-On. "; movies[0].duration = "1 hour 59 minutes.";
    movies[0].reldate = 2024;

    movies[1].name = " Red One. "; movies[1].duration = "2 hours 3 minutes.";
    movies[1].reldate = 2024;

    movies[2].name = " Twisters. "; movies[2].duration = "2 hour 2 minutes.";
    movies[2].reldate = 2024;

    movies[3].name = " Homestead. "; movies[3].duration = "1 hour 52 minutes.";
    movies[3].reldate = 2024;

    movies[4].name = " Abigail. "; movies[4].duration = "1 hour 49 minutes.";
    movies[4].reldate = 2024;

    movies[5].name = " Cuckoo. "; movies[5].duration = "1 hour 42 minutes.";
    movies[5].reldate = 2024;

    movies[6].name = " Trap. "; movies[6].duration = "1 hour 45 minutes.";
    movies[6].reldate = 2024;

    movies[7].name = " Knock at the Cabin. "; movies[7].duration = "1 hour 40 minutes.";
    movies[7].reldate = 2023;

    movies[8].name = " The Substance. "; movies[8].duration = "2 hours 21 minutes.";
    movies[8].reldate = 2024;

```

```
    movies[9].name = " Oppenheimer. "; movies[9].duration = "3 hours.";
    movies[9].reldate = 2023;
```

```
    movies[10].name = " The 4 Corners. "; movies[10].duration = "1 hour 16 minutes.";
    movies[10].reldate = 2022;
```

```
    movies[11].name = " Saltburn. "; movies[11].duration = "2 hours 11 minutes.";
    movies[11].reldate = 2023;
```

```
    Movie* moviepointer = movies;
```

```
    const int rows = 5;
```

```
    const int seatsperrow = 6;
```

```
    char seats[rows][seatsperrow];
```

```
    char (*seatspntr)[seatsperrow] = seats;
```

```
    cout << " You have chosen 1 for Cinema House. " << endl << "\n";
```

```
    do
```

```
    {
```

```
        SetColor(32);
```

```
        cout << setfill('*') << setw(10) << "*";
```

```
        cout << " Cinema House ";
```

```
        cout << setfill('*') << setw(10) << "*" << "\n";
```

```
        SetColor(91);
```

```
        cout << "\n" << " To exit enter any number except the menu numbers. " << endl <<
        "\n";
```



```
SetColor(95);
```

```
cout << " Movie genres :- " << endl;
```

```
cout << " 1. Action Movies. " << endl;
```

```
cout << " 2. Horror Movies. " << endl;
```

```
cout << " 3. Drama Movies. " << endl;
```

```
SetColor(36);
```

```
cout << "\n" << " Enter your choice. ";
```

```
cin >> choice;
```

```
switch (choice)
```

```
{
```

```
case 1:
```

```
    SetColor(91);
```

```
    cout << "\n" << " To exit enter any number except the menu numbers. " << endl  
<< "\n";
```

```
    cout << " You have chosen 1 for Action Movies. " << endl << "\n";
```

```
    SetColor(95);
```

```
    cout << " Menu :- " << endl;
```

```
    for (int i = 0; i < 4; i++)
```

```
    {
```

```
        cout << " " << i + 1 << ". " << (*moviepointer).name << "(" <<  
        (*moviepointer).duration << ")" << " " << (*moviepointer).reldate << ". " << endl;
```

```
        moviepointer++; } }
```

```

SetColor(36);

cout << "\n" << " Enter which Action Movie would you like to watch? ";

cin >> amchoice;


switch (amchoice)
{
case 1:

    cout << " Carry On has been selected. " << endl;

    receipt[0] = movies[0].name;

    receipt[1] = movies[0].duration;

    receipt[2] = to_string(movies[0].reldate);

    break;


case 2:

    cout << " Red One has been selected. " << endl;

    receipt[0] = movies[1].name;

    receipt[1] = movies[1].duration;

    receipt[2] = to_string(movies[1].reldate);

    break;


case 3:

    cout << " Twisters has been selected. " << endl;

    receipt[0] = movies[2].name;

```

```

receipt[1] = movies[2].duration;

receipt[2] = to_string(movies[2].reldate);

break;

```

case 4:

```

cout << " Homestead has been selected. " << endl;

receipt[0] = movies[3].name;

receipt[1] = movies[3].duration;

receipt[2] = to_string(movies[3].reldate);

break;

```

```

default: cout << " Invalid Action Movie choice has been entered. " << endl <<
"\n";

```

```

SetColor(33);

cout << "\n" << "          To exit :- enter -1. " << endl;

cout << "          To continue :- any number. ";

cin >> exit;

cout << "\n";

if (exit == -1)
{
    SetColor(90);

    cout << "\n" << setfill('~') << setw(50) << "~" << endl;

    cout << "    You have exited the Cinema House...  " << endl;

    cout << setfill('~') << setw(50) << "~" << endl << "\n";
}

```

```

        return 0;
    }

    continue;
}

break;

case 2:

    SetColor(91);

    cout << "\n" << " To exit enter any number except the menu numbers. " << endl
    << "\n";

    moviepointer = movies + 4;

    cout << " You have chosen 2 for Horror Movies. " << endl << "\n";

    SetColor(95);

    cout << " Menu :- " << endl;

    for (int i = 4; i < 8; i++)
    {
        cout << " " << i + 1 << ". " << (*moviepointer).name << "(" <<
        (*moviepointer).duration << ") " << (*moviepointer).reldate << ". " << endl;

        moviepointer++;
    }

    SetColor(36);

```

```
cout << "\n" << " Enter which Horror Movie would you like to watch? ";  
cin >> hmchoice;
```

```
switch (hmchoice)
```

```
{
```

```
case 5:
```

```
    cout << " Abigail has been selected. " << endl;
```

```
    receipt[0] = movies[4].name;
```

```
    receipt[1] = movies[4].duration;
```

```
    receipt[2] = to_string(movies[4].reldate);
```

```
    break;
```

```
case 6:
```

```
    cout << " Cuckoo has been selected. " << endl;
```

```
    receipt[0] = movies[5].name;
```

```
    receipt[1] = movies[5].duration;
```

```
    receipt[2] = to_string(movies[5].reldate);
```

```
    break;
```

```
case 7:
```

```
    cout << " Trap has been selected. " << endl;
```

```
    receipt[0] = movies[6].name;
```

```
    receipt[1] = movies[6].duration;
```

```
    receipt[2] = to_string(movies[6].reldate);
```

```
break;
```

case 8:

```
cout << " Knock at the Cabin has been selected. " << endl;
```

```
receipt[0] = movies[7].name;
```

```
receipt[1] = movies[7].duration;
```

```
receipt[2] = to_string(movies[7].reldate);
```

```
break;
```

```
default: cout << " Invalid Horror Movie choice has been entered. " << endl <<
"\n";
```

```
SetColor(33);
```

```
cout << "\n" << "          To exit :- enter -1. " << endl;
```

```
cout << "          To continue :- any number. ";
```

```
cin >> exit;
```

```
cout << "\n";
```

```
if (exit == -1)
```

```
{
```

```
    SetColor(90);
```

```
    cout << "\n" << setfill('~') << setw(50) << "~" << endl;
```

```
    cout << "    You have exited the Cinema House...  " << endl;
```

```
    cout << setfill('~') << setw(50) << "~" << endl << "\n";
```

```
    return 0;
```

```
}
```

```

        continue;
    }

    break;

case 3:

    SetColor(91);

    cout << "\n" << " To exit enter any number except the menu numbers. " << endl
    << "\n";

    moviepointer = movies + 8;

    cout << " You have chosen 3 for Drama Movies. " << endl << "\n";

    SetColor(95);

    cout << " Menu :- " << endl;

    for (int i = 8; i < 12; i++)
    {

        cout << " " << i + 1 << "." << (*moviepointer).name << "(" <<
        (*moviepointer).duration << ")" << (*moviepointer).reldate << "." << endl;

        moviepointer++;
    }

    SetColor(36);

    cout << "\n" << " Enter which Drama Movie would you like to watch? ";

    cin >> dmchoice;

```

```

switch (dmchoice)
{
case 9:
    cout << " The Substance has been selected. " << endl;
    receipt[0] = movies[8].name;
    receipt[1] = movies[8].duration;
    receipt[2] = to_string(movies[8].reldate);
    break;

case 10:
    cout << " Oppenheimer has been selected. " << endl;
    receipt[0] = movies[9].name;
    receipt[1] = movies[9].duration;
    receipt[2] = to_string(movies[9].reldate);
    break;

case 11:
    cout << " The 4 Corners has been selected. " << endl;
    receipt[0] = movies[10].name;
    receipt[1] = movies[10].duration;
    receipt[2] = to_string(movies[10].reldate);
    break;

```


case 12:

```
cout << " Saltburn has been selected. " << endl;

receipt[0] = movies[11].name;

receipt[1] = movies[11].duration;

receipt[2] = to_string(movies[11].reldate);

break;
```

```
default: cout << " Invalid Drama Movie choice has been entered. " << endl <<
"\n";
```

```
SetColor(33);

cout << "\n" << "          To exit :- enter -1. " << endl;

cout << "          To continue :- any number. ";

cin >> exit;

cout << "\n";

if (exit == -1)
{
    SetColor(90);

    cout << "\n" << setfill('~') << setw(50) << "~" << endl;

    cout << "    You have exited the Cinema House...  " << endl;

    cout << setfill('~') << setw(50) << "~" << endl << "\n";

    return 0;
}

continue;
}
```

```

        break;

default: cout << " Invalid Movie Choice has been entered. " << endl;

        SetColor(33);

        cout << "\n" << "                To exit :- enter -1. " << endl;

        cout << "                To continue :- any number. ";

        cin >> exit;

        cout << "\n";

        if (exit == -1)
        {
            SetColor(90);

            cout << "\n" << setfill('~') << setw(50) << "~" << endl;

            cout << "    You have exited the Cinema House...  " << endl;

            cout << setfill('~') << setw(50) << "~" << endl << "\n";

            return 0;

        }

        continue;

    }

    cout << "\n" << " How many tickets would you like to buy? ";

    cin >> tickets;

    receipt[4] = to_string(tickets);

    double total_price = 1200 * tickets;

    // Validate ticket number

```

```

if (tickets <= 0 || tickets > (rows * seatsperrow))
{
    cout << " Invalid number of tickets. Please try again. " << endl << "\n";
    continue;
}

//tickets.

if (tickets > 2)
{
    total_price *= 0.9; // Apply a 10% discount
    cout << " You have received a 10% discount! " << endl << "\n";
}

cout << " Book your seat below now. " << endl << "\n";

for (int i = 0; i < rows; i++)
{
    for (int j = 0; j < seatsperrow; j++)
    {
        seatspntr[i][j] = '0';
    }
}

SetColor(92);

```

```

cout << " Available Seats :- " << endl;

for (int i = 0; i < rows; i++)
{
    for (int j = 0; j < seatsperrow; j++)
    {
        cout << " " << seatspntr[i][j] << " ";
    }

    cout << endl;
}

for (int t = 1; t <= tickets; t++)
{
    SetColor(91);

    cout << "\n" << " Ticket " << t << ": " << endl;

    SetColor(36);

    cout << " Select your seat. " << endl;

    cout << " Enter row number: ";

    cin >> row;

    cout << " Enter column number: ";

    cin >> column;

    if (row > 0 && row <= rows && column > 0 && column <= seatsperrow)
    {
        if (seatspntr[row - 1][column - 1] == '0')

```

```

{
    seatspntr[row - 1][column - 1] = '*';

    cout << " Seat " << t << " booked successfully! " << endl;

}

else

{

    cout << " Seat already booked! Please choose another seat. " << endl << "\n";

    t--; // Retry current ticket

}

}

else

{

    cout << " Invalid seat selection. Please try again. " << endl;

    t--; // Retry current ticket

}

}

```

```

    cout << "\n" << " Total price for " << tickets << " tickets: " << total_price << "
(inclusive of any discounts) " << endl << "\n";

```

```

// Record earnings

```

```

RecordCinemaEarnings(total_price);

```

```

SetColor(92);

```

```

cout << " Available Seats :- " << endl;

for (int i = 0; i < rows; i++)
{
    for (int j = 0; j < seatsperrow; j++)
    {
        cout << " " << seatspntr[i][j] << " ";
    }

    cout << endl;
}

SetColor(31);

if (choice == 1)
{
    cout << "\n" << "\n" << " Your Receipt :- " << endl << "\n";

    for (int i = 0; i <= 5; i++)
    {
        if (i == 0)
        {
            cout << " Name : " << receipt[i] << endl;
        }

        else if (i == 1)
        {

```

```

        cout << " Duration : " << receipt[i] << endl;
    }
    else if (i == 2)
    {
        cout << " Release Date : " << receipt[i] << endl;
    }
    else if (i == 3)
    {
        cout << " Today's Date : " << receipt[i] << endl;
    }
    else if (i == 4)
    {
        cout << " Tickets : " << receipt[i] << endl;
    }
    else if (i == 5)
    {
        cout << " Your Bill : " << total_price << endl;
    }
}

if (choice == 2)
{
    cout << "\n" << "\n" << " Your Receipt :- " << endl << "\n";
}

```

```
for (int i = 0; i < 6; i++)  
{  
    if (i == 0)  
    {  
        cout << " Name : " << receipt[i] << endl;  
    }  
    else if (i == 1)  
    {  
        cout << " Duration : " << receipt[i] << endl;  
    }  
    else if (i == 2)  
    {  
        cout << " Release Date : " << receipt[i] << endl;  
    }  
    else if (i == 3)  
    {  
        cout << " Today's Date : " << receipt[i] << endl;  
    }  
    else if (i == 4)  
    {  
        cout << " Tickets : " << receipt[i] << endl;  
    }  
    else if (i == 5)  
    {
```



```

        cout << " Your Bill : " << total_price << endl;

    }

}

}

if (choice == 3)
{
    cout << "\n" << "\n" << " Your Receipt :- " << endl << "\n";
    for (int i = 0; i < 6; i++)
    {
        if (i == 0)
        {
            cout << " Name : " << receipt[i] << endl;
        }
        else if (i == 1)
        {
            cout << " Duration : " << receipt[i] << endl;
        }
        else if (i == 2)
        {
            cout << " Release Date : " << receipt[i] << endl;
        }
    }
}

```

```

else if (i == 3)
{
    cout << " Today's Date : " << receipt[i] << endl;
}

else if (i == 4)
{
    cout << " Tickets : " << receipt[i] << endl;
}

else if (i == 5)
{
    cout << " Your Bill : " << total_price << endl;
}

}

}

SetColor(33);

cout << "\n" << "          To exit :- enter -1. " << endl;

cout << "          To continue :- any number. ";

cin >> exit;

cout << "\n";

} while (exit != -1);

return 0;

}

```

```

void DisplayTotalEarnings()
{
    const string username = "admin";
    const string password = "1234";
    string inputUsername, inputPassword;
    int operatorChoice = 0;

    cout << "\n" << " Operator Login " << endl << "\n";
    cout << " Username: ";
    cin >> inputUsername;
    cout << " Password: ";
    cin >> inputPassword;

    if (inputUsername == username && inputPassword == password)
    {
        ifstream cinemaFile("CinemaHouseEarnings.txt");
        ifstream coffeeFile("CoffeeShopEarnings.txt");

        double cinemaTotal = 0.0, coffeeTotal = 0.0, value;

        // Cinema House earnings
        if (cinemaFile.is_open())
        {

```

```

while (cinemaFile >> value)
{
    cinemaTotal += value;
}

cinemaFile.close();
}

else
{
    cerr << " Error: Unable to open CinemaHouseEarnings.txt " << endl << "\n";
}

// Coffee Shop earnings
if (coffeeFile.is_open())
{
    while (coffeeFile >> value)
    {
        coffeeTotal += value;
    }

    coffeeFile.close();
}

else
{
    cerr << " Error: Unable to open CoffeeShopEarnings.txt " << endl << "\n";
}

```

```

// total earnings

SetColor(32);

cout << "\n" << setfill('*') << setw(10) << "*";

cout << " Total Earnings. ";

cout << setfill('*') << setw(10) << "*" << "\n";


SetColor(31);

cout << " Cinema House Earnings: RS" << cinemaTotal << endl;

cout << " Coffee Shop Earnings: RS" << coffeeTotal << endl;

cout << " Combined Total Earnings: RS" << (cinemaTotal + coffeeTotal) << endl;

cout << "\n";


SetColor(36);

}

}

//cinema house earning fh
void RecordCinemaEarnings(double ticketPrice)
{
    ofstream outFile("CinemaHouseEarnings.txt", ios::app);

    if (outFile.is_open())
    {
        outFile << ticketPrice << endl;

        outFile.close();
    }
}

```

```

    }

    else

    {

        cerr << " Error: Unable to open CinemaHouseEarnings.txt " << endl << "\n";

    }

}

void RecordCoffeeEarnings(double billamount) {

    ofstream outFile("CoffeeShopEarnings.txt", ios::app);

    if (outFile.is_open())

    {

        outFile << billamount << endl;

        outFile.close();

    }

    else

    {

        cerr << " Error: Unable to open CoffeeShopEarnings.txt " << endl << "\n";

    }

}

// Function to validate membership

bool ValidateMembership()

{

    const string ValidUsername = "member";

    const string ValidPassword = "password123";

```

```

string username, password;

for (int attempts = 1; attempts <= 3; attempts++)
{
    SetColor(36);

    cout << " Enter username: ";

    cin >> username;

    cout << " Enter password: ";

    cin >> password;

    if (username == ValidUsername && password == ValidPassword)
    {
        cout << "\n" << " Access granted. " << endl;

        return true;
    }
    else
    {
        cout << "\n" << " Invalid credentials. Try again. " << endl << "\n";
    }
}

cout << " Access denied. Only Members can play. " << endl << "\n";

return false;
}

```

```

// Snooker game simulation

void PlaySnooker()
{
    SetColor(32);

    cout << "\n" << "    --- Snooker Game --- " << endl << "\n";

    SetColor(36);

    int Player1Score = 0, Player2Score = 0;

    srand(static_cast<unsigned int>(time(0)));

    for (int round = 1; round <= 5; round++)
    {
        cout << "    Round " << round << ": " << endl;

        int p1Points = rand() % 10 + 1; // Random points for player 1

        int p2Points = rand() % 10 + 1; // Random points for player 2

        Player1Score += p1Points;

        Player2Score += p2Points;

        cout << "    Player 1 scores " << p1Points << " points. Total: " << Player1Score <<
        "\n";

        cout << "    Player 2 scores " << p2Points << " points. Total: " << Player2Score <<
        endl << "\n";

    }
}

```



```

cout << "  Final Scores:" << endl;

cout << "  Player 1: " << Player1Score << "\n";

cout << "  Player 2: " << Player2Score << "\n";


if (Player1Score > Player2Score)
{
    cout << "\n" << "  Player 1 wins! " << endl;
}

else if (Player2Score > Player1Score)
{
    cout << "\n" << "  Player 2 wins! " << endl;
}

else
{
    cout << "\n" << "  It's a tie! " << endl;
}
}


// Table Tennis game simulation

void PlayTableTennis()
{
    SetColor(32);

    cout << "\n" << "  --- Table Tennis Game --- " << endl << "\n";

```

```

SetColor(36);

int Player1Score = 0, Player2Score = 0;

srand(static_cast<unsigned int>(time(0)));

while (Player1Score < 11 && Player2Score < 11)
{
    int p1Point = rand() % 2; // Random point for player 1 (0 or 1)
    int p2Point = rand() % 2; // Random point for player 2 (0 or 1)

    if (p1Point > p2Point)
    {
        Player1Score++;
    }
    else if (p2Point > p1Point)
    {
        Player2Score++;
    }

    cout << "    Player 1: " << Player1Score << " | Player 2: " << Player2Score << endl <<
    "\n";
}

cout << "    Final Scores: " << endl;;

```

```

cout << "   Player 1: " << Player1Score << endl;
cout << "   Player 2: " << Player2Score << endl << "\n";

if (Player1Score > Player2Score)
{
    cout << "   Player 1 wins the match! " << endl << "\n";
}
else
{
    cout << "   Player 2 wins the match! " << endl << "\n";
}
}

// Squash game simulation
void PlaySquash()
{
    SetColor(32);

    cout << "\n" << "   --- Squash Game --- " << endl << "\n";

    SetColor(36);

    int Player1Score = 0, Player2Score = 0;

    srand(static_cast<unsigned int>(time(0)));

```

```

while (Player1Score < 21 && Player2Score < 21)
{
    int p1Point = rand() % 2; // Random point for player 1 (0 or 1)
    int p2Point = rand() % 2; // Random point for player 2 (0 or 1)

    if (p1Point > p2Point)
    {
        Player1Score++;
    }
    else if (p2Point > p1Point)
    {
        Player2Score++;
    }

    cout << "    Player 1: " << Player1Score << " | Player 2: " << Player2Score << endl <<
    "\n";
}

cout << "    Final Scores: " << endl;
cout << "    Player 1: " << Player1Score << endl;
cout << "    Player 2: " << Player2Score << endl << "\n";

if (Player1Score > Player2Score)
{
    cout << "    Player 1 wins the match! " << endl << "\n";
}

```

```

else

{
    cout << " Player 2 wins the match! " << endl << "\n";
}

}

// Function to generate a pseudo-random number
int GetRandomNumber(int min, int max)
{
    static bool seedInitialized = false;
    if (!seedInitialized)
    {
        seedInitialized = true;
        time_t t;
        time(&t);
        srand(static_cast<unsigned int>(t));
    }
    return min + (rand() % (max - min + 1));
}

// Function to display the current state of the word
void DisplayWord(const string& word, const bool guessed[])
{
    for (size_t i = 0; i < word.length(); ++i)

```

```

{
    if (guessed[i])
    {
        cout << word[i] << " ";
    }
    else
    {
        cout << "_ ";
    }
}

cout << endl;
}

// Hangman game

void PlayHangman()
{
    string words[] = { "apple", "house", "train", "table", "chair" };
    string ChosenWord = words[GetRandomNumber(0, 4)];

    bool guessed[100] = { false }; // Ensure array size is large enough
    int AttemptsLeft = 6;
    bool WordGuessed = false;

    SetColor(32);

```

```

cout << "\n" << " --- Hangman Game --- " << endl << "\n";

SetColor(36);

while (AttemptsLeft > 0 && !WordGuessed)
{
    cout << "\n" << " Attempts left: " << AttemptsLeft << endl;
    DisplayWord(ChosenWord, guessed);

    cout << " Enter your guess (a single letter): ";
    char guess;
    cin >> guess;

    bool correctGuess = false;
    for (size_t i = 0; i < ChosenWord.length(); i++)
    {
        if (ChosenWord[i] == guess && !guessed[i])
        {
            guessed[i] = true;
            correctGuess = true;
        }
    }
}

```

```

if (!correctGuess)
{
    AttemptsLeft--;

    cout << " Wrong guess! " << endl;
}
else
{
    cout << " Good guess! " << endl;
}

WordGuessed = true;

for (size_t i = 0; i < ChosenWord.length(); i++)
{
    if (!guessed[i])
    {
        WordGuessed = false;

        break;
    }
}

if (WordGuessed)
{
    cout << "\n" << " Congratulations! You've guessed the word: " << ChosenWord <<
endl << "\n";
}

```



```

else

{
    cout << "\n" << " Game over! The word was: " << ChosenWord << endl << "\n";
}

}

// Number guessing game

void PlayNumberGuessingGame()

{
    int NumberToGuess = GetRandomNumber(1, 10);

    int UserGuess = 0;

    int attempts = 5;

    SetColor(32);

    cout << "\n" << " --- Guess the Number Game --- " << endl << "\n";

    SetColor(36);

    cout << " I have picked a number between 1 and 10. Can you guess it? " << endl <<
    "\n";

    while (attempts > 0)
    {
        cout << " Enter your guess: ";

        cin >> UserGuess;
    }

```

```

if (UserGuess < 1 || UserGuess > 10)
{
    cout << " Please enter a number between 1 and 10. " << endl;
    continue;
}

if (UserGuess == NumberToGuess)
{
    cout << " Congratulations! You guessed the number correctly! " << endl << "\n";
    return;
}

else if (UserGuess < NumberToGuess)
{
    cout << " Too low. Try again. " << endl << "\n";
}

else
{
    cout << " Too high. Try again. " << endl << "\n";
}

attempts--;
}

```

```

    cout << " Game over! The correct number was: " << NumberToGuess << endl << "\n";
}

```

```

// Function to display the Tic Tac Toe board

```

```

void DisplayBoard(const char board[9])
{
    cout << "\n";
    for (int i = 0; i < 9; i++)
    {
        cout << " " << board[i] << " ";
        if (i % 3 != 2)
        {
            cout << "|";
        }
        if (i % 3 == 2 && i != 8)
        {
            cout << "\n" << "----+----+----" << "\n";
        }
    }
    cout << "\n";
}

```

```

// Function to check for a winner

```

```

char CheckWinner(const char board[9])
{
    const int WinCombos[8][3] =
    {
        {0, 1, 2}, {3, 4, 5}, {6, 7, 8},
        {0, 3, 6}, {1, 4, 7}, {2, 5, 8},
        {0, 4, 8}, {2, 4, 6}
    };

    for (const auto& combo : WinCombos)
    {
        if (board[combo[0]] != ' ' && board[combo[0]] == board[combo[1]] &&
            board[combo[1]] == board[combo[2]])
        {
            return board[combo[0]];
        }
    }

    return ' ';
}

// Function to check if the board is full
bool IsBoardFull(const char board[9])
{

```

```

for (int i = 0; i < 9; i++)
{
    if (board[i] == ' ')
    {
        return false;
    }
}

return true;
}

// Tic Tac Toe game
void PlayTicTacToe()
{
    char board[9] = { ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ' };
    char currentPlayer = 'X';
    char winner = ' ';

    SetColor(32);
    cout << "\n" << " --- Tic Tac Toe --- " << endl;

    SetColor(36);

    while (winner == ' ' && !IsBoardFull(board))
    {

```

```

DisplayBoard(board);

int move;

cout << "\n" << " Player " << currentPlayer << ", enter your move (1-9): ";
cin >> move;

if (move < 1 || move > 9 || board[move - 1] != ' ')
{
    cout << " Invalid move. Try again. " << endl;
    continue;
}

board[move - 1] = currentPlayer;
winner = CheckWinner(board);

if (winner == ' ')
{
    currentPlayer = (currentPlayer == 'X') ? 'O' : 'X';
}
}

DisplayBoard(board);

```

```

if (winner != ' ')
{
    cout << "\n" << " Congratulations! Player " << winner << " wins! " << endl << "\n";
}
else
{
    cout << "\n" << " It's a draw! " << endl << "\n";
}
}

void RecreoHubMenu()
{
    cout << " Recreohub's Menu :- " << endl;
    cout << " 1. Cinema House. " << endl;
    cout << " 2. Coffee Shop. " << endl;
    cout << " 3. Indoor Games (Only for Members). " << endl;
    cout << " 4. Arcade Games. " << endl;
    cout << " 5. Purchases. " << endl;
    cout << " 6. Exit. " << endl;
}

```